Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 12
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If the mean of $x, x+2, x+4, x+6, x+8$ is 24, then the value of x is: [1]
(A) 20 (B) 22 (C) 24 (D) 18

Q2. In a frequency distribution table, the frequency of the modal class is denoted by f_1. Then f_2 represents the: [1]
(A) frequency of the modal class
(B) frequency of the class succeeding the modal class
(C) frequency of the class preceding the modal class
(D) total frequency

Q3. If $\sum f_i = 50$, $l = 60$, $cf = 15$, $f = 20$, and $h = 10$, the median of the distribution is: [1]
(A) 65 (B) 68 (C) 70 (D) 62.5

Q4. For a moderately skewed frequency distribution, if the median is equal to its mean, then the mode of the distribution is: [1]
(A) equal to the mean (B) greater than the mean
(C) less than the mean (D) zero

Q5. The mean of 100 observations is 50. If one observation 50 was replaced by 150, the resulting mean is: [1]
(A) 50.5 (B) 51.0 (C) 51.5 (D) 52.0

Q6. **Assertion (A):** If the median class is $20-30$ and the modal class is $30-40$, their respective class marks are 25 and 35.
Reason (R): Class mark is given by Class mark $= \dfrac{\text{Lower limit} + \text{Upper limit}}{2}$. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Write the empirical relationship between the three measures of central tendency. Hence express Median in terms of Mean and Mode. [2]

Q8. For the following data, write the modal class and find its class width: [2]

Class	5–10	10–15	15–20	20–25	25–30	30–35
Frequency	6	11	21	23	14	5

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Find the mode of the following distribution: [3]

Class Interval	0–10	10–20	20–30	30–40	40–50
Frequency	8	12	10	11	9

Q10. Find the median of the following distribution of scores: [3]

Class	100–110	110–120	120–130	130–140	140–150
Frequency	6	10	15	13	6

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

Q11. An incomplete distribution having total frequency 230 has median 46. Find the missing frequencies f_1 and f_2: [5]

Class Interval	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	12	30	f_1	65	f_2	25	18

Q12. In a hospital, the ages of 360 patients admitted during a month were recorded. Find both the mean age (using step-deviation method) and modal age: [5]

Age (in years)	5–15	15–25	25–35	35–45	45–55	55–65
Patients	30	60	100	80	50	40

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: STATE POLLUTION CONTROL BOARD CITY AMBIENT AIR QUALITY INDEX (AQI) [4 Marks]

During winter inversion months, environmental scientists recorded the 24-hour peak Air Quality Index (AQI) values across 70 continuous ambient air monitoring stations in a metropolitan region.

AQI Range	50–100	100–150	150–200	200–250	250–300	300–350
Monitoring Stations	5	12	20	18	10	5

(i) Calculate the mode of the Air Quality Index (AQI) recorded across stations. [2]

(ii) Find the median class for the ambient air quality readings. [1]

(iii) How many monitoring stations recorded severe or very poor air (AQI ≥ 200)? [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★